

Key features of hyperbola



1

- a 0 b ∞ c $x = 0$ d $y = 0$
 e $y = x, y = -x$

2

- a -4 b Negative c Positive d 2 and 4

3

- a 0 b $-\infty$ c ∞ d 0
 e 0

4

a

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y	-1	-2	-4	-20	-200	200	20	4	2

- b 0
 c $x = \frac{2}{y}$
 d 0
 e As x approaches 0 from the right, the function value approaches ∞ . As x approaches 0 from the left, the function value approaches $-\infty$.

As x approaches ∞ and $-\infty$, y approaches 0. This is called the limiting value of the function.

5 6

6

- a Yes b No c No

7

- a $x = 0$ b $y = 0$
 c All real values of x , except $x = 0$. d All real values of y , except $y = 0$.

8 The function will be undefined.

9

- a $x = 0$ only b All real values of x , except $x = 0$

c $y = 0$



d All real values of y , except $y = 0$.

- 10 Looking at $y = \frac{2}{x}$, notice that the numerator 2 is non-zero, and so $\frac{2}{x}$ can never be equal to 0.

Another approach is to rearrange the equation and make x the subject. Then we get $x = \frac{2}{y}$.
In this form we can see that the denominator y cannot be equal to 0.

11

a 2

b 1

c $y = \frac{4}{x}$

d The graph moves further from the axes

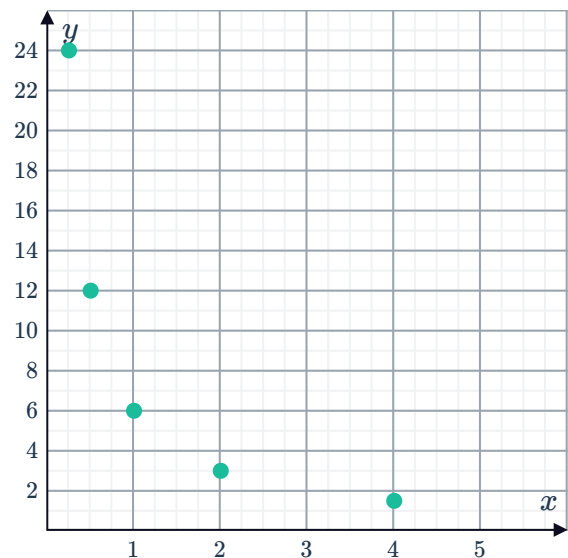
Graphs of hyperbolas

12

a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y	24	12	6	3	$\frac{3}{2}$

b



- 13 She can label a point on the graph.

14

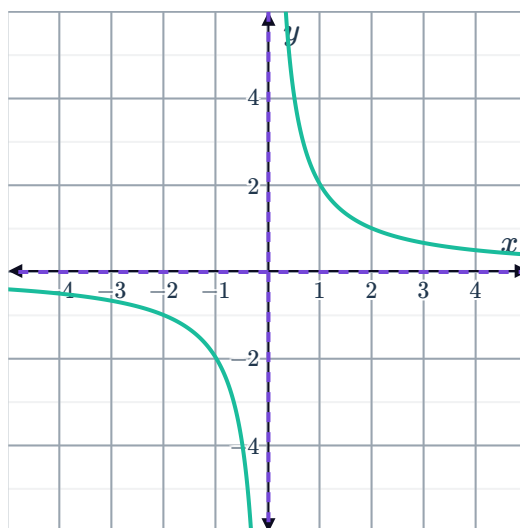


a

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	-1	-2	-4	4	2	1

ii



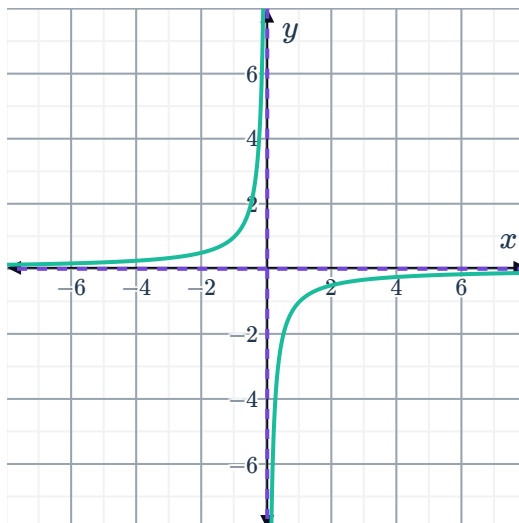
iii 1 and 3

b

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{2}$	1	2	-2	-1	$-\frac{1}{2}$

ii



iii 2 and 4

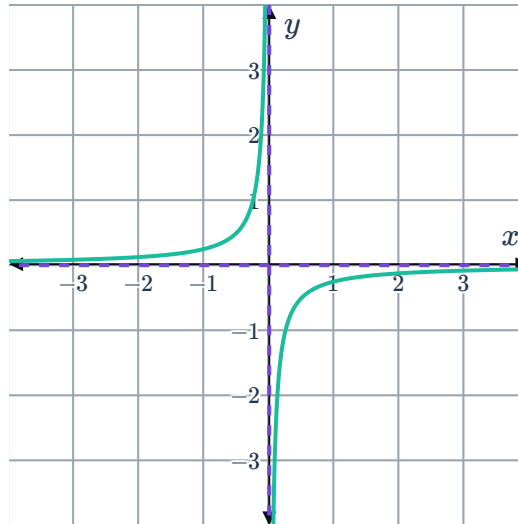
c

i



x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$

ii

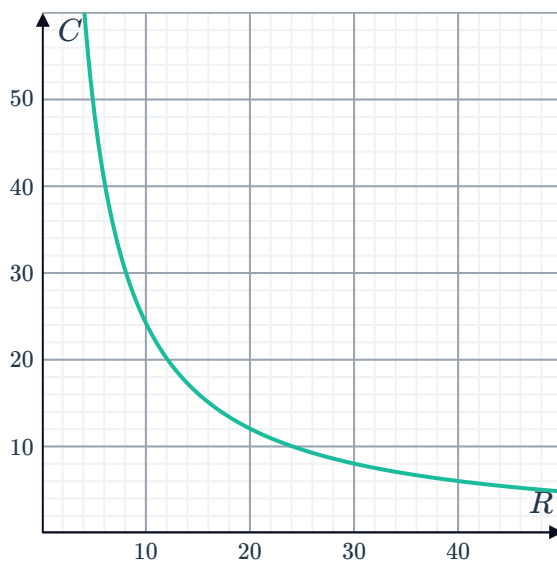


iii 2 and 4

Applications

15

a



b The current decreases.

16



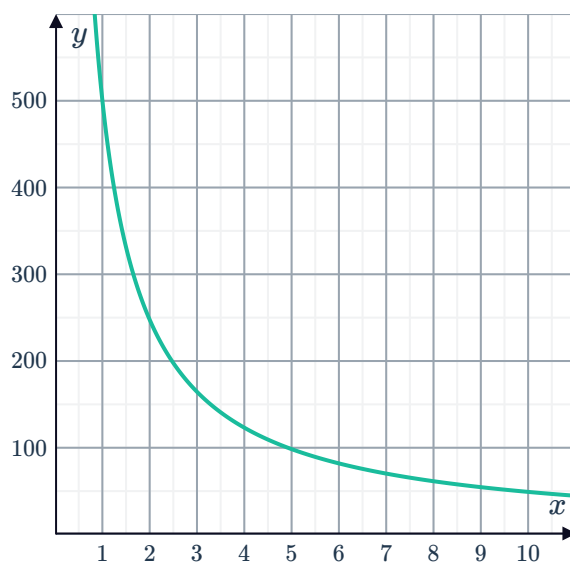
a \$245.00

b $y = \frac{490}{x}$

c

x	1	2	3	4	5	6
y	490	245	163.33	122.5	98	81.67

d



e Inverse variation

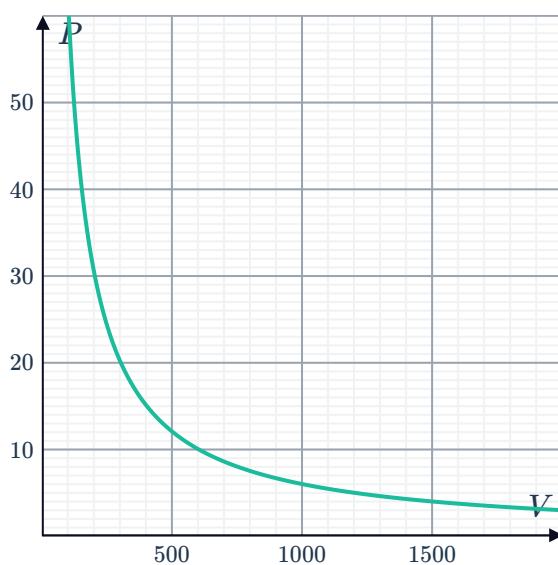
17

a -2

b 32 units²

18

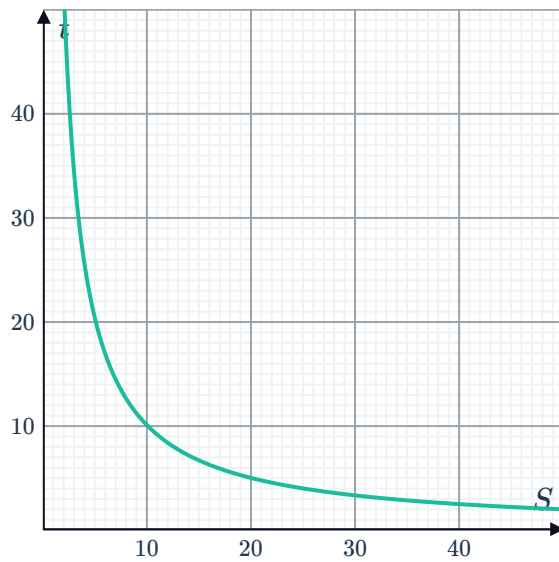
a

b 6000 kg/cm²

c The pressure decreases.

19

a



c 2 hours



b 10 hours

d The speed must increase.